

Action-a-Bot: Exploring Human-Chatbot Conversations for Actionable Instruction Giving and Following

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ABSTRACT

Conversation serves as one critical mechanism for knowledge-sharing and instruction-giving in collaborative work. Conversation allows people to take turns to make contributions, plan joint actions, align shared understanding of work status and resolve action failure. However, when such collaboration involves non-human AI actors like chatbots, there is a lack of understanding of how human participants may respond to the chatbot's prompts and guidance, and whether the interaction can similarly improve the actionability of instructions given to people. In this study, we prototyped a chatbot system, ActionaBot for providing task instructions to novice workers, and conducted an initial study to explore its effects on procedural instruction giving and following. Our results indicate that, novices although might perceive instructions to be inactionable due to prior experience and how instructions were authored, they were able to follow conversational guidance and willing to adapt to the chatbot through turn-taking to calibrate working states back-and-forth. Besides, users' awareness of the work status increased with the conversational prompts from the chatbot.

CCS CONCEPTS

• Human-centered computing → Empirical studies in interaction design.

KEYWORDS

chatbot, instruction-giving, instruction-following, how-to tutorials

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1 INTRODUCTION AND RELATED WORK

Instruction-giving and instruction-following are collaborative in nature, where the knowledge holder and receiver need to work together to accomplish the task. Conversation is one mechanism that people utilize to share and acquire work- and collaboration-related information, such as the state of the joint work (e.g., “which

step are we in?”) and co-workers (e.g., “have you finished that step?”) and the work plan (e.g., “shall we move to the next step?”). Through conversations, team members are able to explicitly coordinate their actions and achieve synergies ([10]).

To perform instruction-giving and -following task through conversations, such as to ask questions, and to obtain guidance required for task execution and completion, numerous mechanisms are in play to make it possible for interlocutors to contribute to work altogether. For example, different team members need to take turns to contribute to the communication to avoid duplication and to allow one person to build upon another person's previous contribution ([19, 21]). Utterances contributed to conversations possess semantic and structural relations to one another for coherent and orderly information exchange ([14–16]). They need to establish and maintain shared knowledge, also known as common ground, for both the conversation and the work per se to unfold smoothly ([22]). CSCW research has allocated much attention to these mechanisms in technology-infused and mediated work ([1]). Theoretical and empirical understandings have been applied to the system design for collaborative processes attended mainly by human actors ([2]). However, it remains an open space in how to apply the learned insights to human-agent collaboration, such as where workers follow the tutorials offered by systems to accomplish tasks.

We present an exploratory study on human-AI conversational collaboration to examine how human workers perform an unfamiliar software installation task under the guidance of an instruction-giving chatbot. Our system, *ActionaBot*, was created to simulate how expert workers may take turns to deliver procedural instructions to novices during the task execution to help them complete the tasks and acquire the task know-hows. ActionaBot aims to improve the ‘actionability’ of regular online tutorials (e.g., tutorials archived as static web pages) by allowing workers to interact with a conversational agent to incrementally attempt and interactively complete the task with heightened situational awareness of the work status, rather than self-exploring the static tutorials based on their limited understanding and awareness of the task.

To explore the feasibility of using a conversational agent to deliver tutorial instructions with enhanced actionability for novices, we prototyped ActionaBot by using the online how-to knowledge sharing platform, wikiHow¹ as the source of tutorial content. WikiHow tutorials are contributed by loosely structured, and volunteering contributors. Such a peer production approach can potentially provide highly rich and diverse content, while at the same time also lacking systematic effort to verify whether the tutorials are comprehensible and/or learnable to inexperienced novices. Besides, novices may lack appropriate meta-cognitive skills to self-monitor

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¹wikiHow platform – <https://www.wikihow.com/wikiHow>

work progress, leading to bigger inactionable barriers during their instruction-following process ([8, 9]). We explore two research questions in this project: 1) Can novice workers follow task instructions provided by an instruction-giving chatbot conversationally in accomplishing a difficult and unfamiliar task? 2) How do novices behave and react in the conversation with the chatbot throughout the work process?

2 CHATBOT-BASED CONVERSATIONAL INSTRUCTION GIVING

2.1 System Overview

ActionaBot includes four interface components (see Figure 1): (1) Chatbot panel, where users type to converse with the bot using free-form text inputs; (2) Task navigation route panel, which visualizes the workflow of the whole task as a directed graph, representing each step as a node in the graph. Nodes are colorized to externalize the chatbot's understanding of current joint work states and used to trace novices' execution; (3) Textual instruction panel, which displays detailed instruction of specific steps; (4) Visual demonstration panel, which shows the supplementary visual illustration corresponding to the textual instruction panel. The instructions and materials are extracted from wikiHow tutorials in verbatim and broken down into a step-by-step format following the markups available in the original document. By delivering the instructions through conversational interactions, novices are asked to focus on a granular piece of sub-procedure. We implemented ActionaBot using a Model-View-Controller software design pattern ([17]) and the ManyChat framework for chatbot building and Flask framework for web-based service integration.

2.2 ActionaBot Design

2.2.1 How-To Tutorials as Knowledge Base. The wikiHow How-To tutorials typically consist of text-based step-by-step instructions and embedded multimedia illustrations, such as images and short video clips. The tutorials are formatted using specific markups ([18, 24]), permitting systematic content export and reformatting. We parsed the markdowns and extracted the instruction of each step in a tutorial and associated media contents.

2.2.2 Dialogue Design. In instruction giving, where an instruction giver tells a worker to perform a task, both parties are required to participate in a joint conversation for presenting their own information, for eliciting co-worker's information, and for building up mutual knowledge (e.g., what one knows and doesn't know), negotiating the work status and next steps, and coordinating mutual actions (cf. [5, 6, 25]). We identified the dialogue acts of ActionaBot in the conversation by borrowing insights gained from observations of human conversational instruction-giving from the literature [3, 4, 11, 20]. To prototype chatbot-based instruction giving, ActionaBot was designed to perform the following dialogue acts: Greeting, Instruction Giving (instruction details), Inquiry (asking novice's execution outcomes, feelings), Suggestion (suggesting novice to review instruction, check the outcome, take next step, go back and review), Request for User's Action, and Encouragement. All possible prompts or responses are domain-independent and

non-task-specific, which can be applied across different areas and tasks to drive the conversation (shown in Appendix Figure 2).

2.2.3 Conversation as a Collaborative Means of Instruction Following. To support collaborative instruction-following through human-chatbot conversations, we make archived tutorials to be conversational. We designed a parsing algorithm to convert the original markup wikiHow tutorial into a sequence of step instructions. The workflow generator generates a directed topological workflow-graph of parsed steps, based on the execution sequence. Parsed step-by-step instructions are then dynamically loaded into ActionaBot on demand (when the novice asks the chatbot to load), supporting the novice conversing with ActionaBot on a small step. Meanwhile, ActionaBot tracks the executed steps, and past steps are colorized according to users' disclosed execution states. As shown in Figure 1, the successfully executed step is colored with green and the step where users have questions is colored with yellow. The color-coding in the procedural graph aims to supplement the conversations. The work trace maintained by ActionaBot is illustrated to support mutual awareness, help people locate potential issues, and keep track of their own work ([12]).

3 PILOT STUDY

We recruited English-speaking volunteers as novice workers to receive instructions from ActionaBot to work on an advanced software installation task. We asked people about their major background, as well as programming and software installation experience. Participants' answers are used as an indicator of their software installation experience levels. People are qualified to participate only when they do not have computer science/engineering related background, limited knowledge in software development and no experience about Python installation. A total of 7 participants (20 to 55 years old; 4 females) from the community of a major US university joined the online study.

The novices were asked to converse with ActionaBot to install Python 3 on a remote computer in real-time. The whole task has three major subprocedures, including downloading TeamViewer on a local computer, connecting the local computer to the remote computer using it, and downloading Python together with testing. To avoid any security concerns, we designed the installation task to perform upon a remote computer, which allowed us to observe how people performed the task directly and controlled the task environment across participants.

Before starting the collaboration task, participants were given an overview of the study process, a tutorial on how to interact with the ActionaBot system interface, and were guided to access the interface. Then, we got users' consent to record study data and proceed with the experiment. During the experiment, the experimenter was fly-on-the-wall ([7]) and took observation notes, with minimum intervention. Participants were asked to share their screens. They were instructed to keep two windows active, one for ActionaBot and the other for task execution. Participants (N=7) took time to finish the instruction-following task. A 30-minute semi-structured interview was then conducted to better understand participants' experiences and thought processes during the study. The post-session interviews were audio-recorded and transcribed using an automated transcriber *Otter* first and then manually verified.

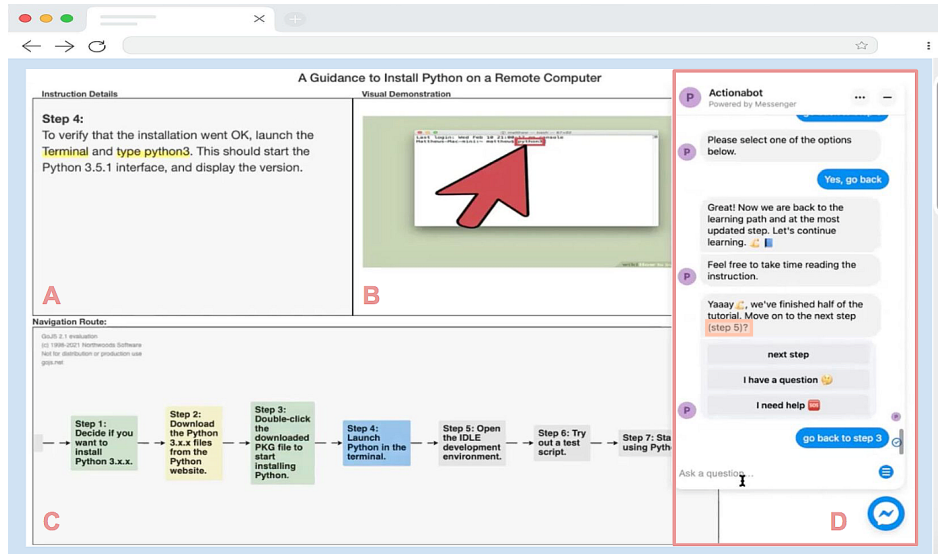


Figure 1: ActionaBot provides novice workers a grounding space to follow the instruction. The main interaction is in panel D, where the user takes a conversation with ActionaBot. The operation cue (shown in the light pink highlight) is provided to map conversation back to the instruction. Panel A shows the textual instruction of the current instruction step, while panel B shows the visual demonstrations of the current instruction step. The content showing in panel A and B are given by ActionaBot based on users' demand. Panel C displays the task navigation route, indicating the workflow of steps and the ActionaBot's understanding of the user's execution status.

4 PRELIMINARY RESULTS

In terms of task completion, only one person got stuck in the first subprocedure and did not finish the task, and the rest six people successfully followed the instructions. Although some participants made mistakes and installed a wrong version, the results suggested that the majority of novice workers could complete the work under the chatbot's conversational guidance. We looked into the interview transcripts and dialogue history to understand how novices interacted with ActionaBot. The transcripts were analyzed using open coding to identify emerging themes in a bottom-up manner. The dialogue history was analyzed under the theoretical foundation of turn-taking ([19]).

4.1 Conversational Following of Step-by-Step Instructions

The dialog history suggests that the participants and ActionaBot indeed took turns to verify the working status of the current step, decided where to pay attention to, and planned on the next action. Most of the novice participants indicated that they were able to follow the instruction and found it helpful to converse with the chatbot to obtain step-wise instruction. As P2 behaved in the conversation, "ActionaBot: Are you confident to finish this step (step 6)? User: Yes, I finished. User: No control yet. " Given how conversational acts were scoped and organized naturally, conversational instruction helped them decompose the whole task into smaller, more actionable units, making it easier to know where to look at and assess the completion status of a step. As P6 explained, "There are no additional things that could have been possibly done (in a

step). ... So I'm pretty sure it's 100% (done). (P6)" Therefore, designing the conversational tutorial helps novices incrementally follow instructions.

However, the participants with differential skills and experiences of computer operation perceived the designated chunks of instruction, which stems from how the original wikiHow materials were authored differently. The participants with less experience (P5, P2) preferred more detailed steps, while the experienced participants (P6) indicated that certain steps could be merged or skipped. While the length and depth of instructional units prescribed by wikiHow could impact actionability by the worker, the exact ways how the impact is manifested would depend on a worker's prior expertise, and potentially how these units were communicated (e.g., merged or skipped by the chatbot) through the conversation.

4.2 Conversational Prompting of Awareness and Actions

Preliminary results indicate the benefits of human-chatbot conversation to increase situational-awareness and situated planning of actions. Four participants reported that having a conversation about the workflow with ActionaBot is helpful since they obtained an awareness of the completion status of a step when the chatbot prompted them to confirm it. As P6 indicated that prompt-driven status confirmation was helpful, "The bot helps, it kind of telling me, like tracking, which step I'm on. (P6)"

Moreover, the chatbot's suggestions of moving on to the next step and revisiting a previous step are beneficial because this design facilitates participants' reflection on the next actionable plan

or potential issues (e.g., self-execution errors or inconsistent instructions) in the prior steps. As stated by P5, she started planning actionable items when receiving the chatbot's prompt to navigate to the next step, *"It's helped me to think of what to do next. (P5)"*

Participants also said the chatbot's suggestions of moving-on, going-back or asking questions is especially useful to solve dilemmas when they got stuck on certain steps, as they were prompted to think about taking actions instead of keeping thinking without doing anything. As stated by P2, *"If it didn't ask me if I wanted to move on. I might be still thinking, like, stuck on the question. And thought, should I move? Or should I not move? (P2)"*

Although ActionaBot prompted novice coworkers to confirm their work status and suggest action to take, some participants indicated that it was difficult to self-assess whether a step is finished. As stated by P7, *"The bot keeps asking me about whether I have already finished it. But it didn't show actually, I don't know, what's the standard of finishes the step. (P7)"* This result implies that displaying existing textual and visual materials from wikiHow may not be sufficient, and novice workers need to be guided to know how to utilize existing information to conduct self-assessment.

4.3 Resolving Discrepancies Around the Conversations

Participants appeared to adapt their behaviors to discrepancies occurring around the conversations by slowing down their execution of task steps to wait for the chatbot's instruction, accelerating to catch up the step the chatbot prompts or negotiating which step to focus on. Discrepancies emerged when work status-oriented information exchanged through the conversation differs from what's displayed visually on the procedural graph or the illustration from the wikiHow tutorial. In human conversations, such discrepancies can lead to communication breakdown and would require a collaborative effort to re-coordinate and repair the conversation. In human-chatbot conversations, however, the chatbot is often technically limited in its ability to collaborate with human counterparts to fix such problems. However, observations from the study indicate participants' willingness to take the agency to "work-around" the technical limitation so that they can continue receiving guidance/benefits by working with the chatbot, as stated by P7, *"The bot sometimes trapped me into a loop. It keeps asking me questions I haven't reached. So I keep on going forward or going backward to check."* It would be beneficial if an instruction-giving chatbot could provide a conversation recovery module to handle discrepancies.

5 FUTURE WORK

This work explores the feasibility of introducing conversation mechanism to archived online tutorials by designing an instruction-giving chatbot, such that novice workers are enabled to accomplish unfamiliar procedural tasks collaboratively with the chatbot. We found that novices are able to be aware of their work status and adjust their behaviors through conversing with our ActionaBot to follow instructions. Through turn-taking, novices and ActionaBot are able to be mutually aware of the work status from the other side and plan on the next action. ActionaBot has the potential to coregulate novice coworkers. Our study also shows the current online tutorials may contain actionability issues, indicating future

design should bridge users to more actionable tutorials. Besides, novices could be of benefit from additional intelligent feedback for self-paced interactive learning when consuming the archived know-hows. In addition, the preliminary results trigger us to reflect whether the step-by-step instruction is detailed enough to support an actionable step or should it be broken down into even smaller increments [13], and whether/how the conversational instruction facilitates situated planning [23].

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APPENDIX

A DETAILS ABOUT THE DIALOGUE ACTS OF ACTIONABOT

Categories	Dialogue Act (ActionaBot)	Description	Utterance Examples
Greeting	Greeting	Greeting when conversation starts	It's nice to meet you!
Statement	Introduction	Introduce the task	We are going to [install python] today...
	Interface Illustration	Tell people about the user interface (information blocks)	The bottom block shows the overall steps of our task.
Question	Users' Outer-State Inquiry	Ask people's execution outcome	Have you finished XX step?
	Users' Inner-State Inquiry	Verify with users about their feeling, confidence, question	Are you confident about the current step? Do you have any doubts?
Suggestion	Situational Work-Flow Suggestion (the instruction flow control)	Suggest users to move forward, go back to previous step	Do you want to move to the next step? Do you want to repeat? Want to jump to the previous step?
	Pre-condition and Outcome Checking Suggestion	Suggest people to compare the pre-condition and outcome with given sources	Can you compare your current status with what is shown in the figure?
Request	Procedure Instruction	Instruct people to execute some step	Let's move to Step XX Let's get started.
Encouragement	Encouragement	Encourage people	Good job!

Figure 2: The categories of system dialogue acts. Since the dialogue acts of ActionaBot are independent to instruction content, the system is generalizable for how-to tutorials on other topics and from other source platforms.